

IA-02 - Questions & Answers

1. Why is Spring Cloud Config Server used?

Answer: It provides centralized configuration for all microservices, including database details, Kafka, Redis, server ports, JPA settings and the JWT secret. This avoids hard-coding these configurations in individual services.

2. Why is Spring Security with JWT used?

Answer: It provides authentication and role-based authorization. A JWT is generated after successful login and is sent with later requests so the services can identify the user and enforce CUSTOMER/ADMIN access.

3. Why is OpenFeign used instead of directly connecting to another service's database?

Answer: OpenFeign provides communication between microservices through REST APIs. The Order Service can request food details from the Food Service and payment processing from the Payment Service without accessing their databases directly.

4. Why is Apache Kafka used?

Answer: Kafka provides asynchronous communication. After a successful order, the Order Service publishes an order-created event, and the Notification Service consumes it instead of being called directly by the Order Service.

5. Why is Redis used?

Answer: Redis is used to cache food details so repeated requests can be served from the cache instead of MySQL. Updating or deleting food also requires cache eviction/update to avoid stale data.

6. Why are Spring Data JPA and MySQL used?

Answer: Spring Data JPA/Hibernate is used for database operations, while MySQL provides persistent storage. Each major microservice maintains its own database/schema.

7. Why is an API Gateway used?

Answer: The API Gateway acts as the common entry point for client requests and routes each request to the appropriate microservice.